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COURSE CODE: ECA0905

COURSE NAME: DIGITAL SIGNAL PROCESSING

PRACTICAL LAB RESULTS

EXP: 5.1:

CODE:

clc; clear; close all;

%%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');

disp(b_rect');

%%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm');

%%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann');

%%% --- Frequency Response Plots ---

figure;

freqz(b_rect,1);

```
title('High-pass FIR using Rectangular Window');
```

```
figure;
```

```
freqz(b_hamm,1);
```

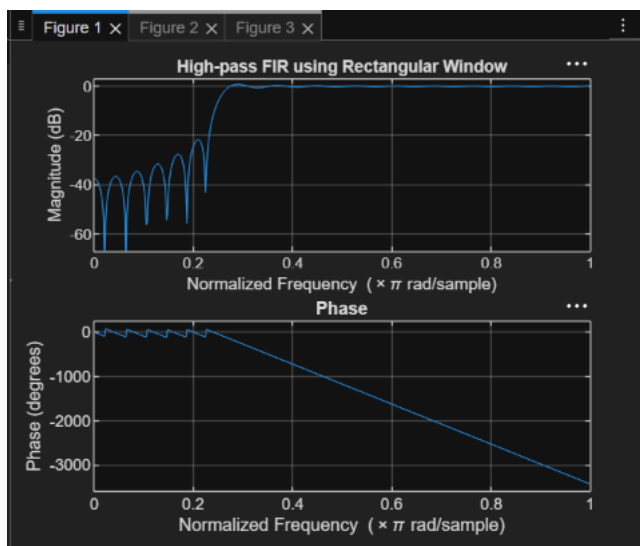
```
title('High-pass FIR using Hamming Window');
```

```
figure;
```

```
freqz(b_hann,1);
```

```
title('High-pass FIR using Hanning Window');
```

OUTPUT:



```
--- High-Pass Coefficients (Rectangular Window) ---  
-0.0089  
  0  
  0.0097  
  0.0143  
  0.0106  
  0  
-0.0117  
-0.0175  
-0.0131  
  0  
  0.0148  
  0.0225  
  0.0171  
  0  
-0.0202  
-0.0314  
-0.0247  
  0  
  0.0318
```

EXP:5.2:

CODE:

```
clc; clear; close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000; % High-pass cutoff frequency (Hz)
```

```
fs = 8000; % Sampling frequency (Hz)
```

```
N = 50; % Filter order
```

```
wc = fp/(fs/2); % Normalized cutoff (0–1)
```

```
%% --- 1. FIR High Pass using Rectangular Window ---
```

```
b_rect = fir1(N, wc, 'high', rectwin(N+1));
```

```
disp('--- High-Pass Coefficients (Rectangular Window) ---');
```

```
disp(b_rect);
```

```
%% --- 2. FIR High Pass using Hamming Window ---
```

```
b_hamm = fir1(N, wc, 'high', hamming(N+1));
```

```
disp('--- High-Pass Coefficients (Hamming Window) ---');
```

```
disp(b_hamm);
```

```
%% --- 3. FIR High Pass using Hanning Window ---
```

```
b_hann = fir1(N, wc, 'high', hanning(N+1));
```

```
disp('--- High-Pass Coefficients (Hanning Window) ---');
```

```
disp(b_hann);
```

```
%% --- Frequency Response Plots ---
```

```
figure;
```

```
freqz(b_rect,1);
```

```
title('High-pass FIR using Rectangular Window');
```

```
figure;
```

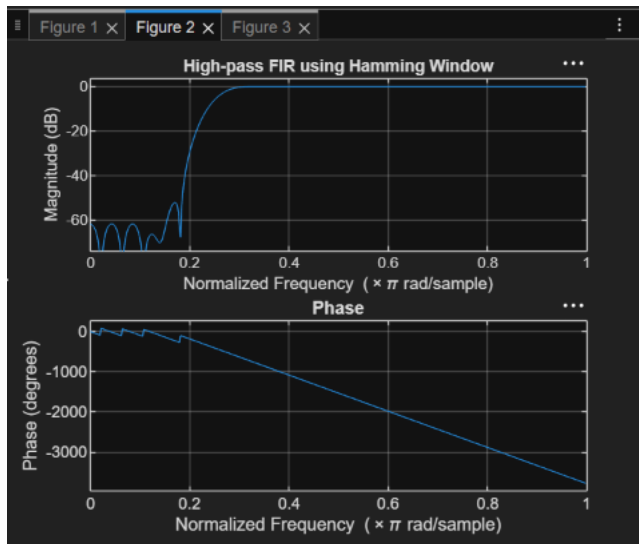
```
freqz(b_hamm,1);
```

```
title('High-pass FIR using Hamming Window');
```

```
figure;
```

```
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT:



```
--- High-Pass Coefficients (Hamming Window) ---
-0.0007
0
0.0009
0.0016
0.0015
0
-0.0024
-0.0044
-0.0039
0
0.0060
0.0103
0.0088
0
-0.0128
-0.0217
-0.0184
0
0.0268
```

EXP:5.3:

CODE:

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect);
```

```

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

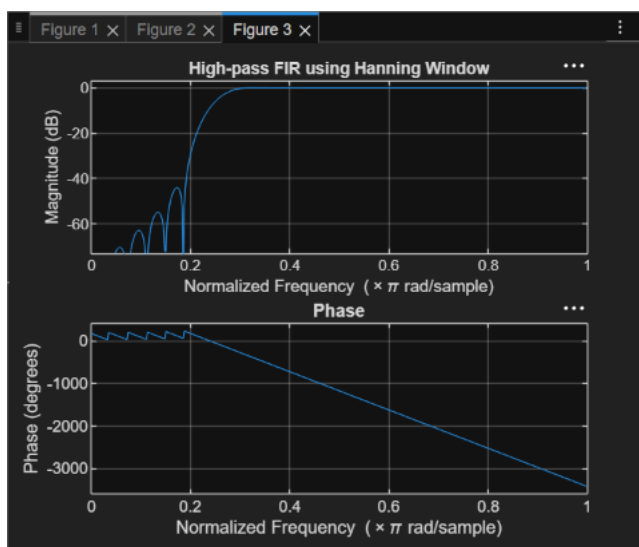
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');

figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

```

OUTPUT:



```

--- High-Pass Coefficients (Hanning Window) ---
-0.0000
0
0.0003
0.0008
0.0009
0
-0.0020
-0.0038
-0.0035
0
0.0057
0.0100
0.0087
0
-0.0127
-0.0216
-0.0183
0
0.0267

```